

CML511 – Quantum Chemistry

5: Harmonic Oscillator

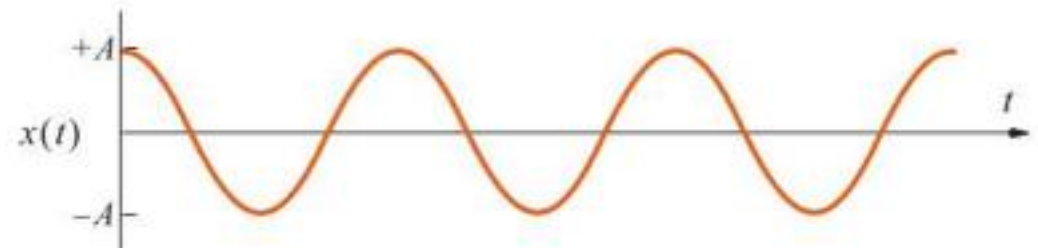
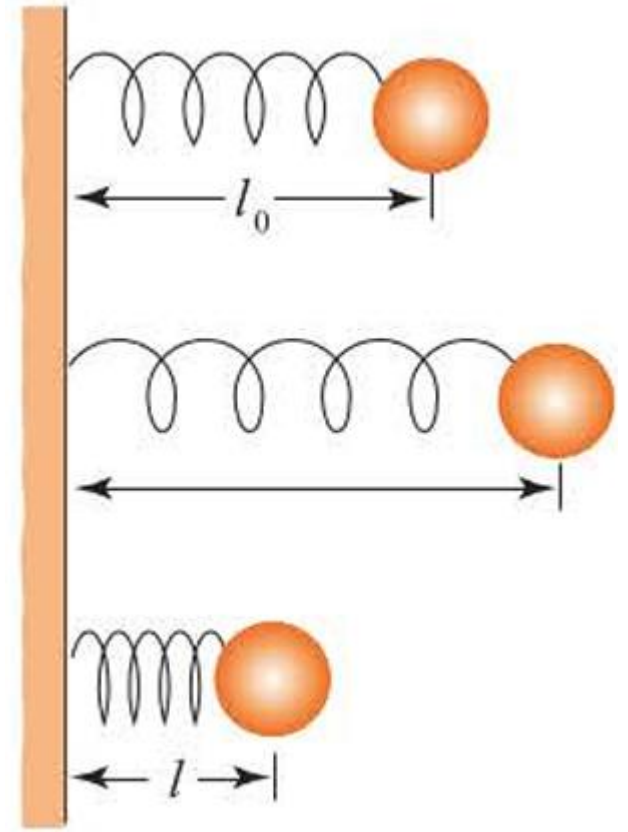
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Vibrations

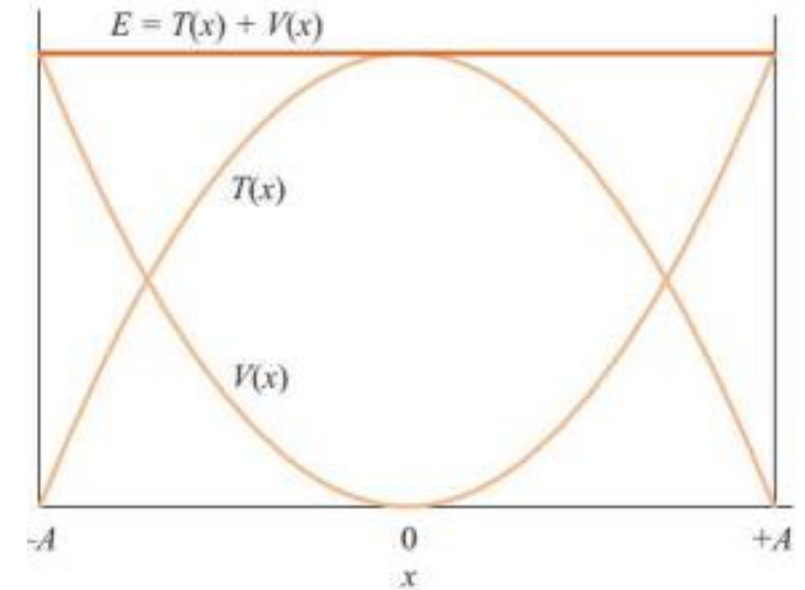
- Hooke's Law: $f = -k(l - l_0) = -kx$
- What does k imply? Why the negative sign?
- Newton's equation: $m \frac{d^2x}{dt^2} + kx = 0$
- General solution: $x(t) = c_1 \sin \omega t + c_2 \cos \omega t$
 $\omega = (k/m)^{1/2}$
- $[\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi]$
- Alternate form: $x(t) = A \sin(\omega t + \phi)$
- Initial displacement $x(0) = c_2 = A$, initial velocity $\left(\frac{dx}{dt}\right)_{t=0} = 0 = c_1 \omega$
- $x(t) = A \cos \omega t$



Energy of Harmonic Oscillator

- $V(x) = -\int f(x)dx + \text{constant} = \frac{1}{2}kx^2 + C = \frac{1}{2}kx^2$ (pot energy is zero for undistorted spring)
- $V(x) = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2 \omega t$
- K.E.: $T = \frac{1}{2}m \left(\frac{dx}{dt}\right)^2 = \frac{1}{2}m\omega^2 A^2 \sin^2 \omega t$
- Total $E = T(x) + V(x) = \frac{1}{2}kA^2$

$$\omega = (k/m)^{1/2}$$



- Total energy is a constant, equal to T_{max} or V_{max}
- This is a **conservative system**

Center of mass

- Diatomic molecule

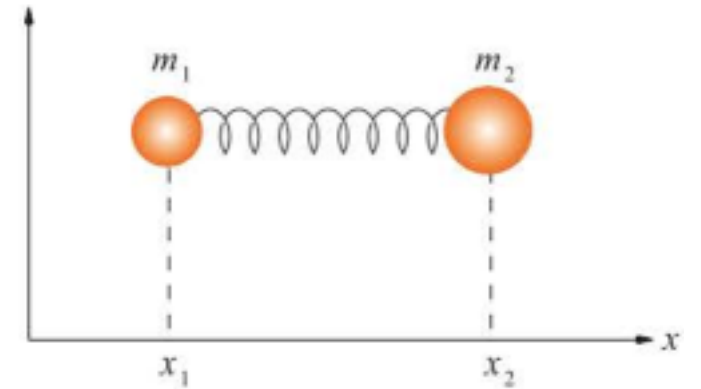
- $m_1 \frac{d^2 x_1}{dt^2} = k(x_2 - x_1 - l_0)$

- $m_2 \frac{d^2 x_2}{dt^2} = -k(x_2 - x_1 - l_0)$

- Add: $\frac{d^2}{dt^2} (m_1 x_1 + m_2 x_2) = 0$

- C.O.M.: $X = \frac{m_1 x_1 + m_2 x_2}{M}$; $M = m_1 + m_2$

- Yields: $M \frac{d^2 X}{dt^2} = 0$. COM moves uniformly with constant momentum

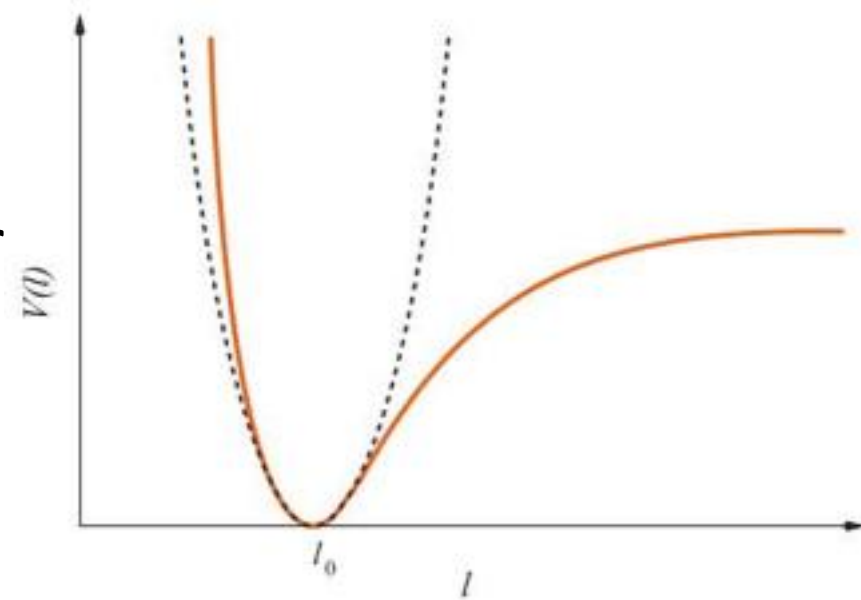


Reduced mass

- Vibration depends on relative coordinate: $x = x_2 - x_1 - l_0$
- $\frac{d^2 x_2}{dt^2} - \frac{d^2 x_1}{dt^2} = -\frac{k}{m_2} (x_2 - x_1 - l_0) - \frac{k}{m_1} (x_2 - x_1 - l_0)$
- $\frac{d^2}{dt^2} (x_2 - x_1) = -k \left(\frac{1}{m_1} + \frac{1}{m_2} \right) (x_2 - x_1 - l_0)$
- $\frac{d^2}{dt^2} (x_2 - x_1 - l_0) = -k \left(\frac{1}{\mu} \right) (x_2 - x_1 - l_0)$
- $\mu \frac{d^2 x}{dt^2} + k x = 0$ Reduced mass
- Using reduced mass, a two – body problem can be treated as a one – body problem.
- If potential energy depends only on rel. distance

Internuclear potential

- Is harmonic oscillator a good approximation for a vibrating diatomic molecule?
- Region around the minimum?
- Taylor expansion



- $$V(l) = V(l_0) + \left(\frac{dV}{dl}\right)_{l=l_0} (l - l_0) + \frac{1}{2!} \left(\frac{d^2V}{dl^2}\right)_{l=l_0} (l - l_0)^2 + \frac{1}{3!} \left(\frac{d^3V}{dl^3}\right)_{l=l_0} (l - l_0)^3 + \dots$$
- 1st term: Zero of energy
- 2nd term: 1st derivative. Zero around the minimum. No force acting between the nuclei $\rightarrow$ equilibrium bond length.
- $$V(x) = \frac{1}{2} kx^2 + \frac{1}{6} \gamma_3 x^3 + \frac{1}{24} \gamma_4 x^4 \dots$$
 Anharmonic terms

Morse potential energy curve

$$V(l) = D_e(1 - e^{-\beta(l-l_0)})^2$$

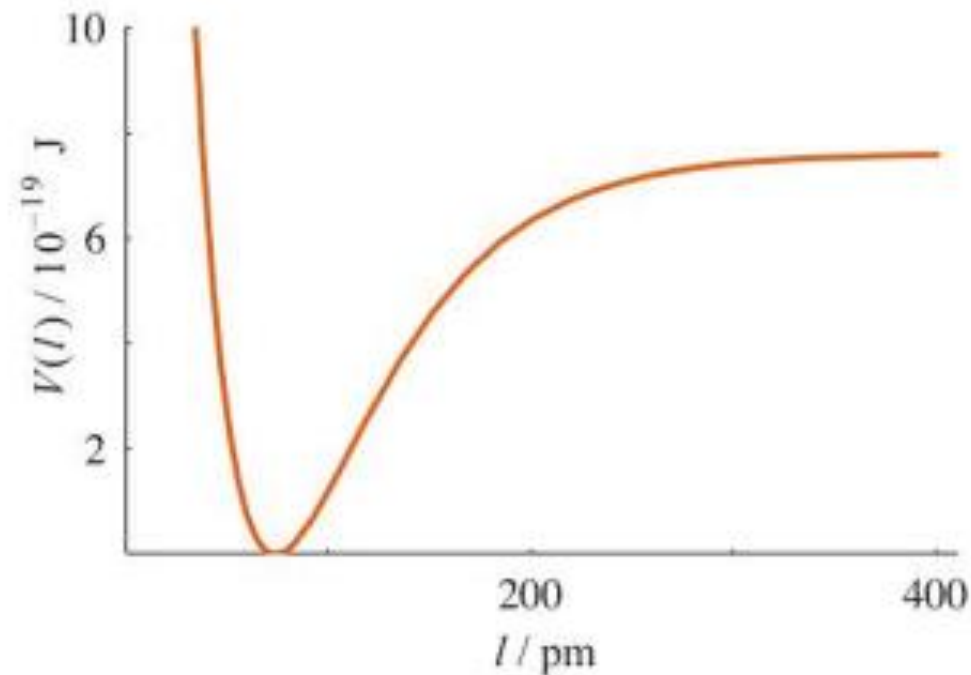


FIGURE 5.6

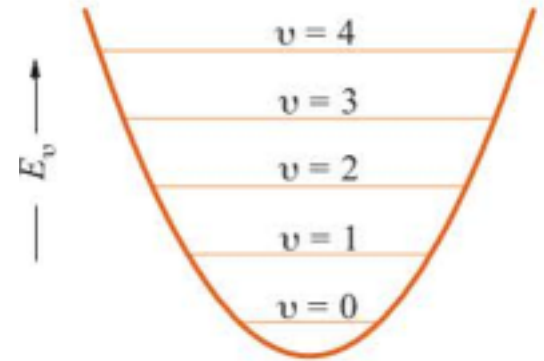
The Morse potential-energy curve $V(l) = D_e(1 - e^{-\beta(l-l_0)})^2$ plotted against the internuclear displacement l for H₂. The values of the parameters for H₂ are $D_e = 7.61 \times 10^{-19}$ J, $\beta = 0.0193 \text{ pm}^{-1}$, and $l_0 = 74.1 \text{ pm}$.

Q.M. Harmonic Oscillator – Energy levels

- TISE:

$$-\frac{\hbar^2}{2\mu} \frac{d^2\psi}{dx^2} + \frac{1}{2} kx^2 \psi(x) = E\psi(x)$$

K.E. P.E.



- Need to solve the second-order differential equation

$$\frac{d^2\psi}{dx^2} + \frac{2\mu}{\hbar^2} \left(E - \frac{1}{2} kx^2 \right) \psi(x) = 0 \quad -\infty < x < \infty$$

- When solved,

$$E_v = h\nu \left(v + \frac{1}{2} \right) \quad v = 0, 1, 2, \dots \quad \nu = \frac{1}{2\pi} \left(\frac{k}{\mu} \right)^{1/2}$$

- Uniform spacing, zero point energy – uncertainty principle

Infrared spectrum

- $E_v = h\nu \left(v + \frac{1}{2} \right) \quad v = 0, 1, 2, \dots$
- Transition (absorption): $\Delta E = E_{v+1} - E_v = h\nu_{obs}$
- Selection rule: $\Delta v = \pm 1$ (+1 absorption, -1 emission)
- Frequency: $\nu_{obs} = \frac{1}{2\pi} \left(\frac{k}{\mu} \right)^{1/2} \text{ Hz}$ (Typically 10^{14} Hz – IR region)
- In wavenumbers: $\tilde{\omega} = \frac{1}{2\pi c} \left(\frac{k}{\mu} \right)^{1/2} \text{ cm}^{-1}$
- Determine force constants of bonds, e.g. H^{35}Cl , $\tilde{\omega}_{obs} = 2886 \text{ cm}^{-1}$

TABLE 5.1

Vibrational Spectroscopic Parameters of Some Diatomic Molecules

Molecule	$\tilde{\omega}_e/\text{cm}^{-1}$	$\tilde{x}_e\tilde{\omega}_e/\text{cm}^{-1}$ ^b	$k/\text{N}\cdot\text{m}^{-1}$	$\bar{l}(v=0)/\text{pm}$ ^a	$D_0/\text{kJ}\cdot\text{mol}^{-1}$ ^b
H ₂	4403.56	123.86	510	74.14	432.1
D ₂	3116.33	62.51	527	74.15	439.6
H ¹⁹ F	4138.32	89.88	920	91.68	566.2
H ³⁵ Cl	2990.94	52.819	478	127.46	427.8
H ⁷⁹ Br	2648.97	45.218	381	141.44	362.6
H ¹²⁷ I	2309.01	39.644	291	160.92	294.7
¹² C ¹⁶ O	2169.81	13.288	1857	112.83	1070.2
¹⁴ N ¹⁶ O	1904.20	14.075	1550	115.08	626.8
¹⁴ N ¹⁴ N	2358.57	14.324	2243	109.77	941.6
¹⁶ O ¹⁶ O	1580.19	11.98	1142	120.75	493.6
¹⁹ F ¹⁹ F	916.64	11.236	454	141.19	154.6
³⁵ Cl ³⁵ Cl	559.72	2.675	319	198.79	239.2
⁷⁹ Br ⁷⁹ Br	325.321	1.0774	240	228.11	190.1
¹²⁷ I ¹²⁷ I	214.502	0.6147	170	266.63	148.8
²³ Na ²³ Na	159.125	0.7255	17	307.89	69.5
³⁹ K ³⁹ K	92.021	0.2829	9.7	390.51	49.6

a. The quantity $\bar{l}(v=0)$ is the average internuclear separation in the $v=0$ state.b. The meanings of $\tilde{x}_e\tilde{\omega}_e$ and D_0 are explained in Section 5.7.

Overtone and Anharmonicity

- $V(x) = \frac{1}{2}kx^2 + \frac{1}{6}\gamma_3x^3 + \frac{1}{24}\gamma_4x^4 \dots$
- Harmonic oscillator predicts only one line in the spectrum, but there are weaker lines at **almost integral multiples** of the **fundamental** line – **overtone**s.
- $\tilde{\omega}_{obs} = \tilde{\omega}_e v - \tilde{x}_e \tilde{\omega}_e v(v+1) \quad v = 1, 2, \dots$

TABLE 5.2
The Vibrational Spectrum of H³⁵Cl

Transition	$\tilde{\omega}_{obs}/\text{cm}^{-1}$	$\tilde{\omega}_{obs}/\text{cm}^{-1}$		$\tilde{\omega}(0 \rightarrow v) / v\tilde{\omega}(0 \rightarrow 1)$
		Harmonic oscillator $\tilde{\omega} = 2885.90v$	Anharmonic oscillator $\tilde{\omega} = 2988.90v - 51.60v(v+1)$	
0 → 1 (fundamental)	2885.9	2885.9	2885.7	1.000
0 → 2 (first overtone)	5668.0	5771.8	5668.2	0.982
0 → 3 (second overtone)	8347.0	8657.7	8347.5	0.964
0 → 4 (third overtone)	10 923.1	11 543.6	10 923.6	0.946
0 → 5 (fourth overtone)	13 396.5	14 429.5	13 396.5	0.928

- $\tilde{\omega}_{obs} = \tilde{\omega}_e \nu - \tilde{x}_e \tilde{\omega}_e \nu(\nu + 1) \quad \nu = 1, 2, \dots$

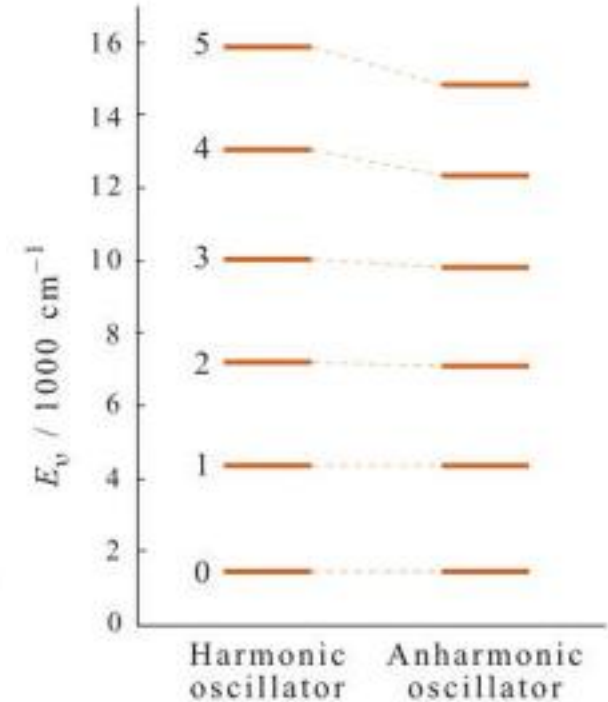
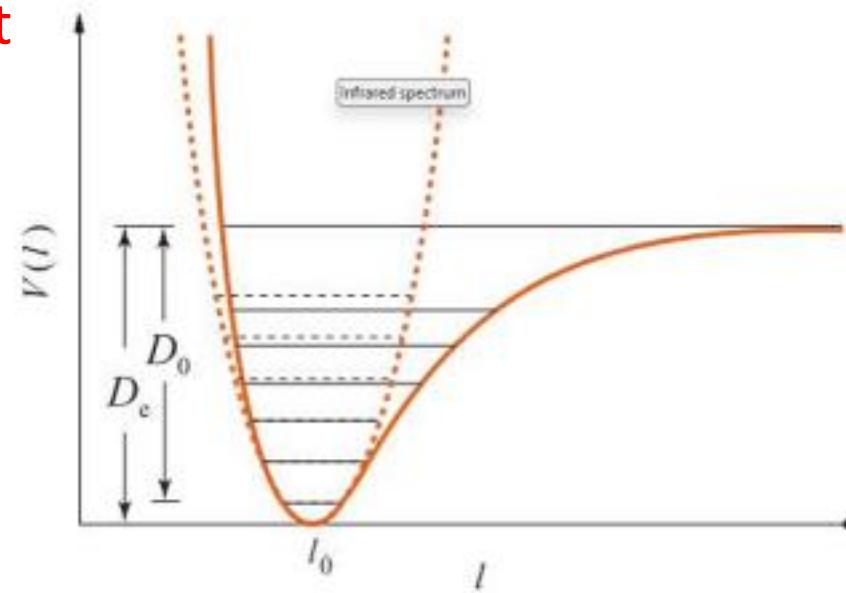
Anharmonicity constant
 $\tilde{x}_e \ll 1$

- Dissociation:

- $D_0 = D_e - \frac{1}{2} h\nu$

- Room temperature

- $k_B T \sim 25 \text{ meV} \sim 200 \text{ cm}^{-1}$
- Only the first few levels are important at r.t.



Wave functions

$$\psi_v(x) = |v\rangle = N_v H_v(\alpha^{1/2} x) e^{-\alpha x^2/2}$$

$$\alpha = \left(\frac{k\mu}{\hbar^2}\right)^{1/2} \quad N_v = \frac{1}{(2^v v!)^{1/2}} \left(\frac{\alpha}{\pi}\right)^{1/4}$$

Hermite polynomials

$$H_v(\alpha^{1/2} x)$$

TABLE 5.3

The First Few Hermite Polynomials ^a

$H_0(\xi) = 1$	$H_1(\xi) = 2\xi$
$H_2(\xi) = 4\xi^2 - 2$	$H_3(\xi) = 8\xi^3 - 12\xi$
$H_4(\xi) = 16\xi^4 - 48\xi^2 + 12$	$H_5(\xi) = 32\xi^5 - 160\xi^3 + 120\xi$

a. The variable ξ is equal to $\alpha^{1/2} x$, where $\alpha = (k\mu)^{1/2}/\hbar$.

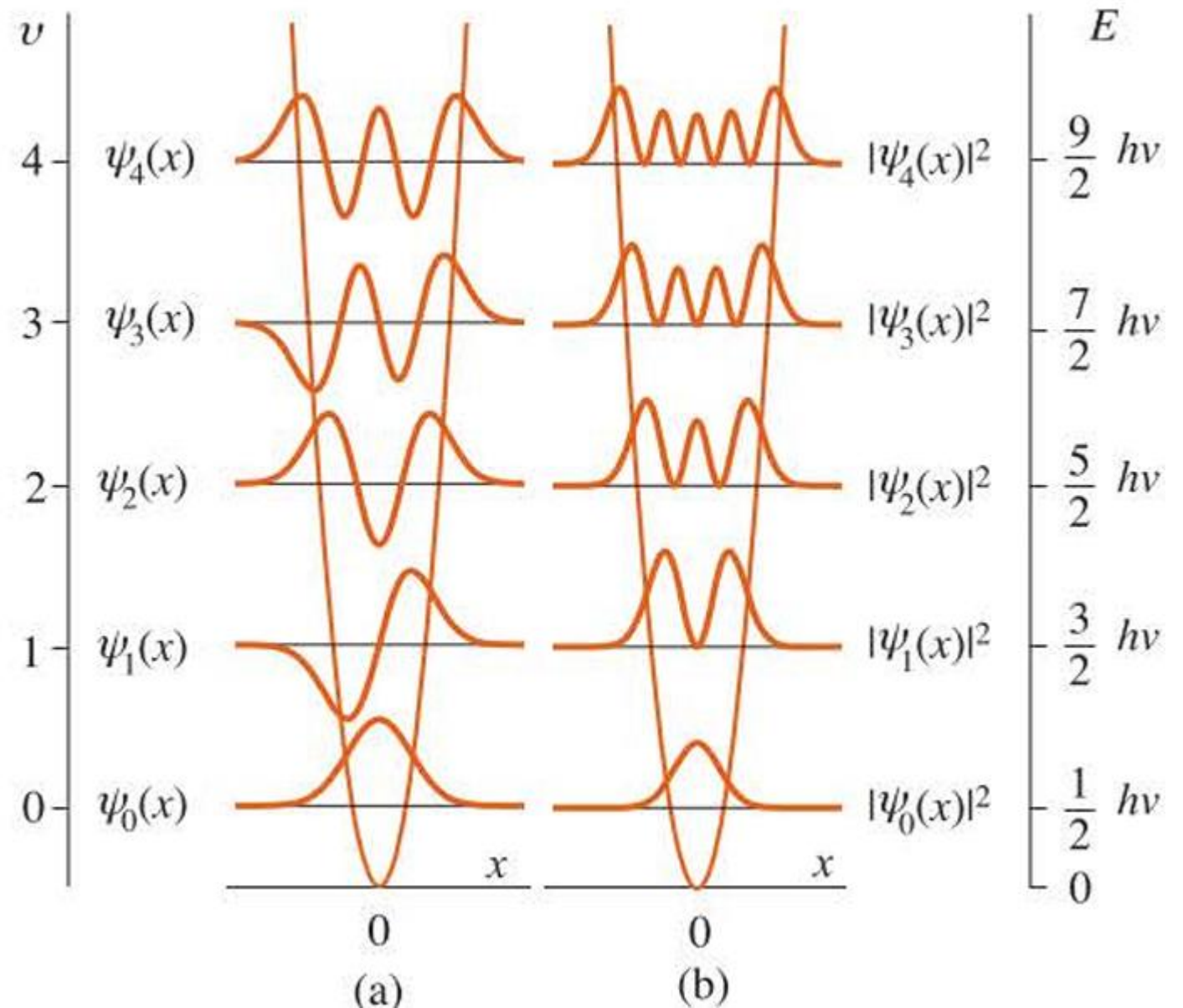
Wave functions

$$\psi_0(x) = |0\rangle = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

$$\psi_1(x) = |1\rangle = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\alpha x^2/2}$$

$$\psi_2(x) = |2\rangle = \left(\frac{\alpha}{4\pi}\right)^{1/4} (2\alpha x^2 - 1) e^{-\alpha x^2/2}$$

$$\psi_3(x) = |3\rangle = \left(\frac{\alpha^3}{9\pi}\right)^{1/4} (2\alpha x^3 - 3x) e^{-\alpha x^2/2}$$



Verify Solutions

- $\psi_0(x) = |0\rangle = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$
- Substitute in SE: $\frac{d^2\psi_0}{dx^2} + \frac{2\mu}{\hbar^2} \left(E_0 - \frac{1}{2}kx^2\right) \psi_0(x) = 0$
- $\left(\frac{\alpha}{\pi}\right)^{1/4} (\alpha^2 x^2 e^{-\alpha x^2/2} - \alpha e^{-\alpha x^2/2}) + \frac{2\mu}{\hbar^2} \left(\frac{\hbar\omega}{2} - \frac{kx^2}{2}\right) \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2} \stackrel{?}{=} 0$
- $(\alpha^2 x^2 - \alpha) + \left(\frac{\mu\omega}{\hbar} - \frac{\mu k}{\hbar^2} x^2\right) \stackrel{?}{=} 0$ where $\alpha = \left(\frac{k\mu}{\hbar^2}\right)^{1/2}$; $\omega = \left(\frac{k}{\mu}\right)^{1/2}$
- ψ_0 is a solution

Orthogonal wave functions

$$\int_{-\infty}^{\infty} dx \psi_v(x) \psi_{v'}(x) = \langle v | v' \rangle = 0; \quad v \neq v'$$

$$\int_{-\infty}^{\infty} dx H_v(\alpha^{1/2}x) H_{v'}(\alpha^{1/2}x) e^{-\alpha x^2} = \int_{-\infty}^{\infty} d\xi H_v(\xi) H_{v'}(\xi) e^{-\xi^2} = 0; \quad v \neq v'$$

Hermite polynomials are orthogonal w.r.t. weighting function $e^{-\xi^2}$

Odd/even functions

- Even function $f(x) = f(-x)$.
Symmetric when reflected across y-axis
- Odd function $f(x) = -f(-x)$. Changes sign upon reflection across y-axis.
- $x^2, e^{-x^2}, \cos x, \cosh x$ even
- $x^3, \sin x, xe^{-x^2}$ odd
- $e^x, x + x^2, \ln x, \sin x + \cos x$ neither odd, nor even
- Hermite polynomials are odd/even

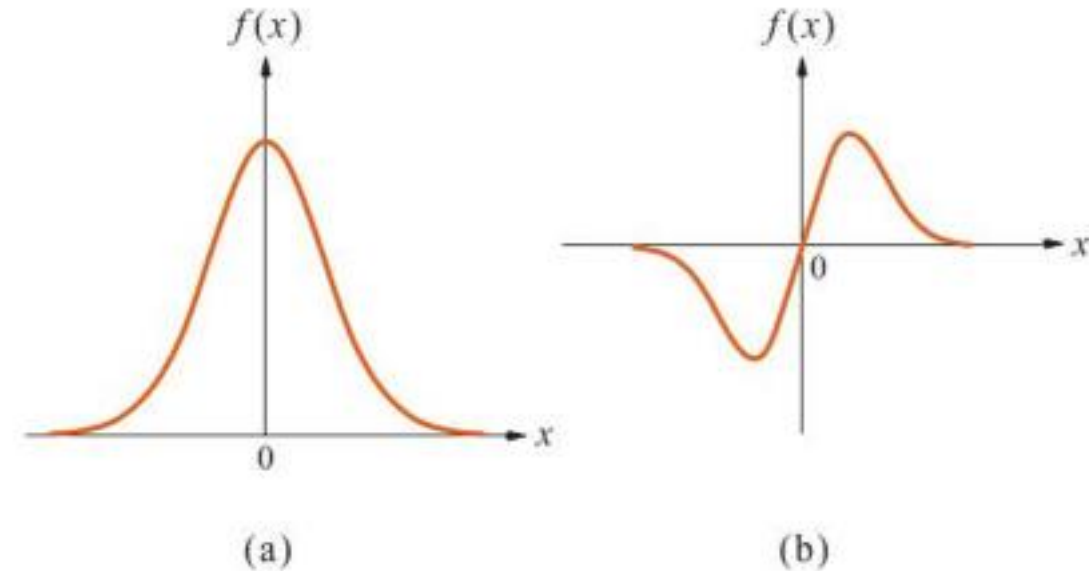


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a. The variable ξ is equal to $\alpha^{1/2}x$, where $\alpha = (k\mu)^{1/2}/\hbar$.

Odd/Even

$$\int_{-a}^{+a} f(x) dx = 0 \quad f(x) \text{ odd}$$

- Average displacement of H.O.: $\langle v | x | v \rangle = 0$
- Average momentum: $\langle p \rangle = \langle v | \hat{p}_x | v \rangle = 0$

$$\int_{-a}^{+a} f(x) dx = 2 \int_0^a dx f(x) \quad f(x) \text{ even}$$

Hermite polynomials

- Recursion formula:

$$H_{v+1}(\xi) - 2\xi H_v(\xi) + 2vH_{v-1}(\xi) = 0$$

- Use recursion formula to show $\langle \xi \rangle = \int_{-\infty}^{\infty} \psi_v(\xi) \xi \psi_v(\xi) d\xi$
- Evaluate $\langle \xi^2 \rangle, \langle p \rangle, \langle p^2 \rangle$

Ladder Operator Method

- $H\psi = \frac{1}{2\mu} [p^2 + (\mu\omega x)^2]\psi = E\psi$
- Factor the Hamiltonian: $u^2 + v^2 = (iu + v)(-iu + v)$, but these are not numbers!! Still!!
- $a_{\pm} = \frac{1}{\sqrt{2\hbar\mu\omega}} (\mp ip + \mu\omega x)$
- $a_- a_+ = \frac{1}{2\hbar\mu\omega} (ip + \mu\omega x)(-ip + \mu\omega x)$
- $= \frac{1}{2\hbar\mu\omega} [p^2 + (\mu\omega x)^2 - i\mu\omega(xp - px)]$
- $= \frac{1}{2\hbar\mu\omega} [p^2 + (\mu\omega x)^2] - \frac{i}{2\hbar} [x, p] = \frac{1}{2\hbar\mu\omega} [p^2 + (\mu\omega x)^2] - \frac{i}{2\hbar} (i\hbar)$
- $= \frac{1}{2\hbar\mu\omega} [p^2 + (\mu\omega x)^2] + \frac{1}{2} = \frac{1}{\hbar\omega} H + \frac{1}{2}$
- $\hat{H} = \hbar\omega \left(a_- a_+ - \frac{1}{2} \right)$ and $\hat{H} = \hbar\omega \left(a_+ a_- + \frac{1}{2} \right)$
- $a_+ a_- = \frac{1}{\hbar\omega} H - \frac{1}{2}$ and so, $[a_-, a_+] = 1$

Raising and Lowering

- TISE: $H\psi = \hbar\omega \left(a_{\pm} a_{\mp} \pm \frac{1}{2} \right) \psi = E\psi$
- Claim: If ψ satisfies TISE with energy E , then $a_+\psi$ satisfies TISE with energy $(E + \hbar\omega)$
- Proof: $H(a_+\psi) = \hbar\omega \left(a_+ a_- + \frac{1}{2} \right) (a_+\psi) = \hbar\omega \left(a_+ a_- a_+ + \frac{1}{2} a_+ \right) \psi$
- $= \hbar\omega a_+ \left(a_- a_+ + \frac{1}{2} \right) \psi = a_+ \left[\hbar\omega \left(a_+ a_- + 1 + \frac{1}{2} \right) \psi \right]$
- $= a_+ (H + \hbar\omega) \psi = a_+ (E + \hbar\omega) \psi = (E + \hbar\omega) (a_+ \psi)$
- Likewise: $H(a_-\psi) = \hbar\omega \left(a_- a_+ - \frac{1}{2} \right) (a_-\psi) = \hbar\omega a_- \left(a_+ a_- - \frac{1}{2} \right) \psi$
- $= a_- \left[\hbar\omega \left(a_- a_+ - 1 - \frac{1}{2} \right) \psi \right] = a_- (H - \hbar\omega) \psi$
- $= a_- (E - \hbar\omega) \psi = (E - \hbar\omega) (a_- \psi)$

Ground state ψ_0

- Lowest rung: $a_- \psi_0 = 0$
- $\frac{1}{\sqrt{2\hbar\mu\omega}} \left(\hbar \frac{d}{dx} + \mu\omega x \right) \psi_0 = 0$ or $\frac{d\psi_0}{dx} = -\frac{\mu\omega}{\hbar} x \psi_0$
- Solve: $\int \frac{d\psi_0}{\psi_0} = -\frac{\mu\omega}{\hbar} \int x dx \Rightarrow \ln \psi_0 = -\frac{\mu\omega}{2\hbar} x^2 + \text{constant}$
- i.e. $\psi_0(x) = A e^{-\frac{\mu\omega}{2\hbar} x^2} = \left(\frac{\mu\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{\mu\omega}{2\hbar} x^2}$ (Normalize - assignment)
- Put in TISE: $\hbar\omega \left(a_+ a_- + \frac{1}{2} \right) \psi_0 = E_0 \psi_0$
- Since, $a_- \psi_0 = 0$, $E_0 = \frac{1}{2} \hbar\omega$
- Raising operator: $\psi_v(x) = A_v (a_+)^v \psi_0(x)$ with $E_v = \left(v + \frac{1}{2} \right) \hbar\omega$
- First excited state: Assignment!!